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# An Exploration of Measures of Spread in High Dimensional Latent Spaces

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## Abstract

Understanding geometric properties of natural language processing models’ latent spaces allows the manipulation of these properties for improved performance on downstream tasks. One such property is the amount of data spread in a model’s latent space, or how fully the available latent space is being used. In this work, we define data spread, and we demonstrate that the commonly used measures of data spread, *Average Cosine Similarity* and a partition function min/max ratio  $I(V)$ , do not provide reliable metrics to compare the use of latent space across models. We propose and examine eight alternative measures of data spread, all but one of which improve over these current metrics when applied to seven synthetic data distributions. Of our proposed measures, we recommend one principal component-based measure and one entropy-based measure that provide reliable, relative measures of spread and can be used to compare models of different sizes and dimensionalities.

## 1 Introduction

The product of many neural network models is a representation of the data input in a high dimensional latent space. The distribution of data in this latent space is often used in the application of the learned model through data clustering for classification, measuring distance between data points to quantify similarity, generation of synthetic data elements, or any number of other downstream tasks. For this reason, understanding and manipulating the geometric properties of models’ latent spaces is an area of active research.

One such geometric property is the question of how evenly the data is distributed in its latent representation. It has been shown that many common neural network architectures produce highly anisotropic latent spaces (1; 2; 3), with data representations taking up only a small portion of the high-dimensional latent space. It has been suggested that a more complete utilization of the latent space could increase the expressiveness of latent data representations. Indeed, various methods for enforcing more complete use of a model’s latent space have been shown to improve performance on benchmarking tasks (4; 5; 6; 7).

However, the majority of this work quantifies the spread of data in a latent space with two measures of isotropy that we argue are inherently flawed. The first common measure of isotropy is the average cosine similarity between data points in the latent space, and the second is a measure introduced by Mu et al. (7) based on a ratio of principal component loadings. First we argue that isotropy as a concept only describes one aspect of an ideally filled space in any number of dimensions. Additionally, while both of these measures can be used to demonstrate that a latent space is absolutely isotropic, they fail to provide a reliable measure for comparing the degree of isotropy present between data in two latent spaces. We present several synthetic distributions in latent space and show that these measures do not behave as would be expected of a reliable measure of relative data spread.

We examine several alternative ways to quantify data spread and compare these measures’ performance on the same example distributions in latent space. We consider two principal component measures as well as the percent overlap of Euclidean distances between the first and  $k$ th nearest neighbor. We also consider several entropy-based measures, including two variants of differential entropy and several measures of relative entropy (Kullback-Leibler Divergence). We show that all proposed measures, except for the KNN distance overlap, behave more intuitively on our evaluation distributions than the two commonly used measures. In particular, we find that the *Eigenvalue Early Enrichment* score and the *Vasicek Ratio MSE* produce scores that most closely mirror our expectations across these example distributions and various numbers of dimensions. Finally, we investigate the behavior of the best-performing of the alternative data spread metrics on real (not simulated) latent spaces produced by a pre-trained Word2Vec model.

## 2 Related Work

In a natural language processing context, using continuous vector representations of language data provides a much more flexible and expressive latent landscape for the model to learn than does the use of hard-coded features, such as term frequency or syntactic and semantic labels (8). Theoretically, with more dimensions available in this continuous latent space, the model can express more finely grained information up until there are enough dimensions to enable memorization of the training data. However, this increased expressiveness as a function of latent space dimensionality depends on the model effectively using (or even being capable of using) the space across all of these dimensions.

This assumption of an evenly used latent space is not guaranteed, as addressed by a growing body of work that assesses and manipulates the geometry of natural language processing models’ latent spaces. Mimno and Thompson (3) explore several geometric characteristics of the Word2Vec Skipgram model’s latent space, notably finding that there is a large positive inner product between the global mean vector across all words and individual word vectors. Similarly, Mu et al. (7) show that common static word embedding models each have a non-zero global mean vector, and that the explained variance of principal components in each model’s latent space decays nearly exponentially. Considering distribution of data representations in common contextual natural language processing models, Ethayarajh (1) find a non-zero average cosine similarity between word vectors throughout all layers of ELMo, BERT, and GPT-2.

Gao et al. (9) suggest that this incomplete use of space, which they term “representation degeneration”, is due to a combination of these models’ maximum likelihood function and the Zipfian distribution of word frequency, which causes most word vectors to be moved together as “negative” examples much more often than they are moved apart when seen in the raw text data. Word frequency has also been shown to be directly correlated with word vector magnitude (10) and distance traveled during training (11). Mu et al. (7) find that the top two PCA components of static language model latent spaces are largely dedicated to expressing word frequency information.

Attempts to “correct” this incomplete use of the latent space include transforming or fine-tuning a fully trained natural language processing model, as well as regularizing during the training process to manipulate the latent space geometry. Post-training adjustments include subtracting the global mean vector and removing highly explanatory principal components (7; 6; 12), using an autoencoder framework towards a similar goal (5), and learning a transformation into a more uniformly filled non-Euclidean space (13). Adjustments made during training include minimizing an additional loss function (2; 9; 14) and using an isotropic Gaussian prior in a VAE (15) or during batch normalization (16). All methods listed above have resulted in improvement on common natural language processing

benchmarking tasks, suggesting that using the latent space more completely may in fact increase the expressiveness of a natural language processing model.

Beyond improving benchmark performance, using the latent space more fully has also been a focus in some work towards more efficient quantization methods. Quantization is a data compression technique in which continuous data is mapped to a set of discrete values (such as 0/1 or the integers). If data representations are using the latent space in a more uniform manner, the data becomes more separable and more easily quantized (17). Based on the methods of Mu et al. (7) described above, Liao et al. (18) remove top principal components during an iterative quantization process. Sablayrolles et al. (17) instead use a nearest-neighbor entropy approximation (Kozachenko-Leonenko entropy) to enforce uniform spread before quantizing the data. Both of these methods find less data loss after compression than quantization methods that do not focus on using the latent space more completely.

### 3 Data Spread in High Dimensions

While the idea of increasing the expressiveness of a model by producing data representations that maximally use the latent space seems fairly intuitive, the details around what it means to “maximally use the latent space” are far from obvious. Add to this the complication of working in high dimensional space, which is anything but intuitive, and defining an ideal latent space according to this theory of maximized expressiveness becomes a research question in itself. Here we explore existing concepts and distributions that can be used to help define this ideal space, as well as some geometric properties of high dimensional space that come up repeatedly in the rest of this work.

#### 3.1 Data Spread vs. Isotropy

One concept that comes up in much of the literature in Section 2 is that of isotropy, which is common across many mathematical and scientific fields. The definition of isotropy varies from field to field, but broadly it is defined as “[i]dentical in all directions; invariant with respect to direction” (19). In the context of data representations in a latent space, this means that the distribution of data representations would be the same in any direction from the origin. Perhaps a more specifically applicable definition comes from statistics: Rudelson (20) finds that a probability distribution is in isotropic position if its covariance matrix is the identity. Similarly, Zhou et al. (16) define a latent space as isotropic if all dimensions have the same variance and are uncorrelated.

Rotational invariance and uncorrelated dimensions are certainly relevant to the idea of a “filled” latent space, but with these definitions, isotropy seems insufficient to define our ideal latent space. While a measure of isotropy can tell us whether our data are distributed similarly in all available directions, it says nothing about what the distribution is in any given direction. For example, data distributed on the shell of a hypersphere are equally isotropic to data distributed in a multivariate normal distribution. Intuitively we’d expect those two data distributions to perform differently on a measure of how fully the available space is being used.

Here we have introduced the term *isotropic* as meaning that space appears to be the same in all directions (“all dimensions have the same variance and are uncorrelated”). In this manuscript, we will use the term *spread* to expand on this notion; a distribution shows desirable *spread* if it evenly spreads points in all directions (isotropy) as well as along each axis/direction.

#### 3.2 Reference Distributions

In considering a latent space that is maximally filled, an obvious choice for how data should be distributed is according to a uniform distribution. By definition, the uniform distribution will maximally fill any shape in any number of dimensions, however this requires a defined shape with hard outer limits. In other words, data in a uniform distribution will maximally fill a hypersphere of any radius, but the probability abruptly drops to zero outside of that radius. This may not make sense for data representations in a continuous latent space.

Alternatively, we might consider a multivariate normal distribution ideal for maximally filling a latent space. Like a uniformly filled hypersphere, data in a multivariate normal distribution will be isotropic. However, the normal distribution does not suffer from the same issue of having a hard cutoff in probability, instead having probability approach zero gradually as distance from the origin increases.

Although using a multivariate normal distribution will result in having a higher density of points around the origin than we would find in referencing a uniform distribution, this gradual decrease in probability (and point density) generally seems a realistic and desirable trait for data representations in a latent space.

### 3.3 Geometry

While not central to our work, there are several non-intuitive geometric characteristics that will come up when discussing data spread in high dimensional spaces. Here we provide a brief description of two particular characteristics that are often included in defining the so-called “Curse of Dimensionality”.

First, the volume and surface area of a hypersphere (with a fixed radius) approach zero as the number of dimensions grows above  $\sim 7$ . Second, the volume approaches zero more quickly than the surface area, which results in almost all of the points within a uniformly filled hypersphere being concentrated on a very thin shell of that hypersphere. The equations for surface area ( $S_d(r)$ ) and volume ( $V_d(r)$ ) in  $d$  dimensions are shown, respectively, in Equations 1 and 2 below.

$$S_d(r) = \frac{2\pi^{d/2}}{\Gamma(d/2)} r^{d-1} \quad (1)$$

$$V_d(r) = S_d \frac{r}{d} \quad (2)$$

In Equation 1 the gamma function in the denominator will dominate as  $d$  grows, bringing the surface area to zero as the number of dimensions increases. The additional  $d$  factor in the denominator of the volume function causes  $V_d(r)$  to approach zero even more quickly than  $S_d(r)$  as  $d$  grows, with the result that data points are increasingly concentrated in the neighborhood of a hypersphere shell with a fixed radius as the number of dimensions grows <sup>1</sup>.

## 4 Models for Comparison

In order to compare different ways of measuring data spread in high dimensional space, we sought a variety of example distributions such that we can assess whether a metric is behaving in a manner that agrees with our notion of even distribution across space. While we could use the actual output of popular natural language processing models, we don’t necessarily know or have strong intuition about how fully these models’ data representations fill the latent space or how they should compare to each other on a sufficient measure of data spread. Instead, we developed a collection of seven structured distributions, each with  $d = \{2, 10, 50, 100\}$  dimensions, and used these as intuitive benchmarks on our proposed measures of spread. Our example latent spaces each have  $250d$  data points.

For spherical distributions, we created a *Shell*, a *Nested Shell*, a uniformly filled *Sphere*, and a *Cone*. For cluster-based distributions, we created clusters that are symmetric across the origin and identical in size (*Symmetric Clusters*), asymmetric clusters of identical size (*Shifted Clusters*), and symmetric clusters of unequal size (*Uneven Clusters*). Details of the characteristics of these distributions follow.

### 4.1 Distribution Definitions

#### Shell

A relatively simple way to sample points from the surface of a  $d$ -dimensional unit hypersphere is to sample from a multivariate standard normal and normalize the length of the associated vectors (21). Here,  $X_i^{shell}$  is the vector for one of the  $250d$  points sampled from our  $d$ -dimensional shell.

$$X_i^{shell} = \frac{X_i}{\|X_i\|}, \quad X_i \sim N(0, I_d) \quad (3)$$

<sup>1</sup>It should be noted that this effect is only observed when considering a bounded space, such as a hypersphere, as can be seen by comparing the histograms of the vector norms of a hypersphere and a normal distribution in Figure 3.

### Nested Shell

To create a nested shell, that is, a shell within a shell, we adjust the radius of half of our shell points to be  $\frac{1}{2}$  instead of 1.

$$r_i^{nest} = \begin{cases} 1 & i \leq 250d/2 \\ 1/2 & i > 250d/2 \end{cases} \quad (4)$$

$$X_i^{nest} = r_i^{nest} X_i^{shell} \quad (5)$$

### Sphere

We can sample from a uniformly filled unit hypersphere by taking the points from our *Shell* distribution and randomizing their distance from the origin between 0 and the radius of the hypersphere,  $r$ . However, due to the exponential relationship between the radius and the volume of a  $d$ -dimensional hypersphere (as seen in Equations 1 and 2,  $V_d = f(r^d)$ ), we cannot sample the distance of each point directly from a  $U(0, r)$  distribution without causing points to be more densely concentrated around the origin. Thus, we invert this exponent when sampling our distance from the origin, as seen in Equation 6.

$$l_i^{sphere} = l_i^{\frac{1}{d}}, \quad l_i \in L \sim U(0, r) \quad (6)$$

$$X_i^{sphere} = l_i^{sphere} X_i^{shell} \quad (7)$$

### Cone

A 3-dimensional cone can be described as a continuous series of circles (2-dimensional spheres), stacked along a third dimension (the cone's height), where the radius of each circle is a function of the distance from the origin,  $l$ , and the angle/width of the cone,  $\theta$ : as we move further from the origin (as  $l$  grows), the radius of each circle increases according to the width of the cone.

Analogously, a  $d$ -dimensional hypercone can be imagined as a series of  $(d - 1)$ -dimensional hyperspheres that are continuously stacked along the  $d$ th dimension. Again, the radius of the  $i$ th stacked sphere,  $r_i^{stack}$ , is a function of  $l_i^{cone}$  (the sphere's distance from the origin) and  $\theta$  (the angle/width of the cone)<sup>3</sup>. As in the case of the *Sphere* distribution, we invert the exponential relationship between volume and distance from the origin to uniformly sample within the cone as in Equation 8.

$$l_i^{cone} = l_i^{\frac{1}{d}}, \quad l_i \in L \sim U(0, r) \quad (8)$$

$$r_i^{stack} = l_i^{cone} \tan(\theta) \quad (9)$$

We then sample from an  $(d - 1)$ -dimensional sphere of radius  $r_i^{stack}$  as in Equations 6 and 7. The points sampled from these  $(d - 1)$ -dimensional spheres ( $X_i^{stack}$ ) are concatenated with the distance from the origin ( $l_i^{cone}$ ) to produce  $d$ -dimensional vectors as in Equation 10

$$X_i^{cone} = (l_i^{cone}, X_i^{stack}) \quad (10)$$

### Clusters

Cluster distributions were created by randomly sampling  $d$  cluster centers, mirroring these centers over the origin to create a total of  $2d$  cluster centers, and randomly sampling around each center  $\mu_j$  to create clusters. In Equation 12 we use the floor function to ensure integer division so that the *Symmetric Clusters* distribution sampled 125 points around each cluster center.

$$\mu_j^{symm} = \begin{cases} \mu_j & j \leq d \\ -\mu_{j/2} & j > d \end{cases}, \quad \mu_j \sim U(-1, 1) \quad (11)$$

$$X_i^{symm} \sim N(\mu_{\lfloor i/125 \rfloor}^{symm}, \min(1/d, 0.2)) \quad (12)$$

<sup>2</sup>This exponential relationship is directly related to the discussion in Section 3.3, since points that are distributed uniformly within a high-dimensional hypersphere will end up largely in the neighborhood of the hypersphere's radius.

<sup>3</sup>We chose to define  $\theta = 1/\sqrt{d}$  as it produced distributions that more consistently demonstrated the strengths and weaknesses of common and proposed measures of spread across different numbers of dimensions.

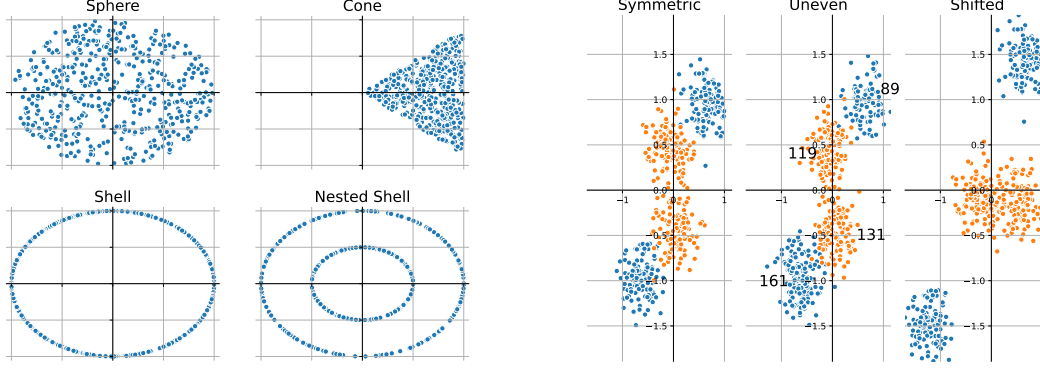


Figure 1: Example distributions visualized in 2 dimensions.

To break symmetry, we amend Equation 12 in two ways. For the *Shifted Clusters* distribution, we randomly shifted each cluster center and again sampled 125 points around each cluster center.

$$\mu_i^{shift} = \mu_i^{symm} + S_i, \quad S_i \sim U(0, 1) \quad (13)$$

For the *Uneven Clusters* distribution, we randomized the number of points in each symmetric cluster, such that the clusters in each mirrored cluster pair,  $(\mu_i^{symm}, \mu_{i+d}^{symm})$ , have  $k$  and  $(250 - k)$  points, respectively, and  $k$  is sampled from a  $U(0, 250)$  distribution.

## 4.2 Visualizing the Distributions

Figure 1 shows all seven distributions in 2-dimensional space, and provides a visual basis for our intuitions about how the distributions compare in terms of how fully they use the space. In these images, the *Sphere* distribution most closely approaches a “full” latent space, perhaps followed by the *Nested Shell* and then the *Shell* distributions (which are at least uniform in all directions). In comparison, the *Cone* and cluster-based distributions do not fill the available space evenly.

However, as discussed in Section 3.3, our ability to visualize data spread declines quickly as the number of dimensions increases, and the intuitions based on these 2-dimensional visualizations become similarly weak. For this reason, we have included two other ways of visualizing our example distributions to help regain some of that intuition.

Figure 2, shows density plots of the raw data values along each dimension in the 10-dimensional distributions. Even in just ten dimensions, the distribution of data among the three spherical distributions (*Shell*, *Nested Shell*, *Sphere*) are quite similar, and even the *Cone* distribution is fairly similar with the exception of the  $d$ th dimension along which the cone points.

The density plots for the cluster distributions help to demonstrate the effect of the transformations between the *Symmetric Clusters* distribution and the *Shifted* and *Uneven Clusters* distributions. They also make clear the stark difference between the data uniformity of the spherical distributions and the cluster based distributions.

Figure 3 shows histograms of vector norms for our *Sphere*, *Cone*, and *Symmetric Clusters* distributions in 2, 10, and 100 dimensions, with a reference normal distribution for comparison<sup>4</sup>. The first item of note is that, as the number of dimensions grows, the shape of the *Sphere* and *Cone* histograms becomes increasingly similar to the expectation for the *Shell* histogram (in which all vectors have the same norm by definition). This follows from the characteristic described in Section 3.3, in that points within a hypersphere are forced onto a thin shell of that hypersphere in high dimensions.

Our cluster distributions are not affected by this characteristic of high-dimensional geometry, since each cluster is defined by a mean and variance, rather than a strict radius. This is apparent in the

<sup>4</sup>The *Cone* distribution’s vector norms are longer than the other distributions across all dimension counts. This is an artifact of holding the variance equal across all distributions, which we do to provide a more accurate comparison to the normal distribution as described in Section 6.3

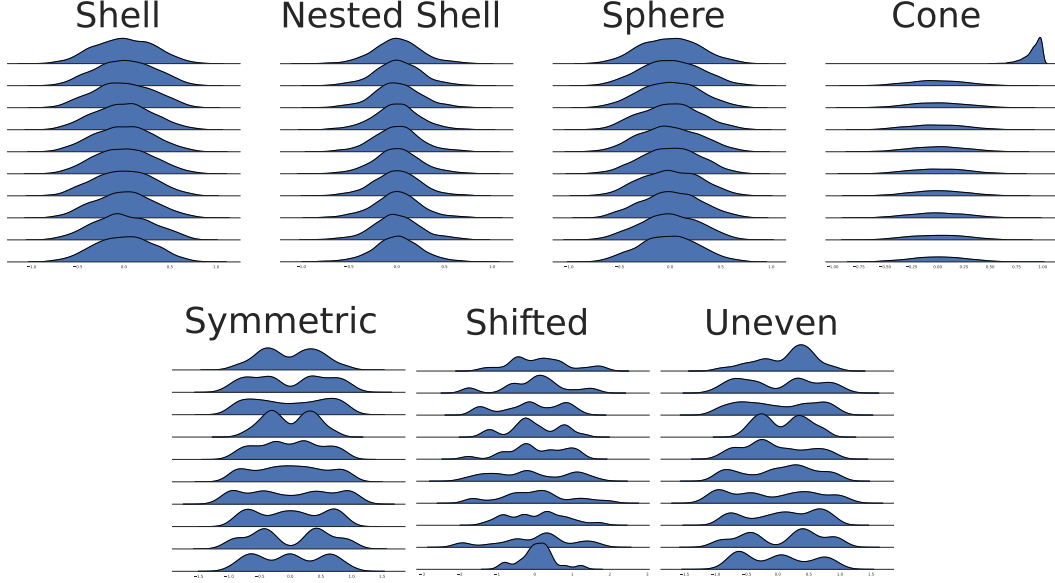


Figure 2: Density plots for 10-Dimensional example distributions show distribution of data values along each dimension; Top row shows the four spherical models; Bottom row shows the four cluster models.

histograms for the *Symmetric Clusters* distribution, which is largely unchanged in shape as the dimensions grow.

After considering these visualizations of our example distributions in higher dimensions, our intuitions about how our models should perform on relative measures of data spread are as follows:

1. The *Shell*, *Nested Shell*, and *Sphere* distributions should all have similar scores that indicate well-spread distributions, particularly in high dimensions.
2. The *Cone* distribution should have a slightly lower score that still indicates a fairly well-spread distribution, particularly in high dimensions.
3. All three cluster models should have similar scores that indicate poorly spread distributions, regardless of the dimensionality of the distribution.

## 5 Common Measures of Isotropy

Research into the effect that data spread has on natural language processing model performance generally relies on one or two measures of spread: Average Cosine Similarity and  $I(V)$ . We argue that both of these common measures fall short as relative measures of spread in high dimensional space.

### 5.1 Average Cosine Similarity

The first and simpler of these two common measures is the average pairwise cosine similarity between word representations in a natural language processing model (4; 1; 22; 9; 6). In an isotropic latent space, the expected pairwise cosine similarity between data points is zero. Note that this measure considers only the angle between two word representations, and not the magnitude of those representations; this means that it can not reflect the extent of in-fill within a space, for example remaining oblivious to the difference between the *Sphere* distribution and either of the shell distributions.

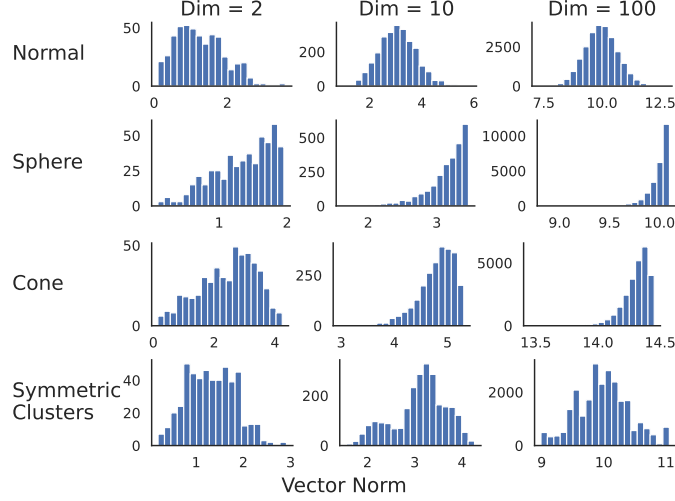


Figure 3: Histograms of vector norms for our reference *normal* distribution and our *Sphere*, *Cone*, and *Symmetric Clusters* example distributions in 2, 10, and 100 dimensions

## 5.2 $I(V)$

The second commonly used measure of spread,  $I(V)$  (sometimes called  $\gamma$ ), is a min/max ratio of a principal component-based partition function.  $I(V)$  was first introduced by Mu et al. (7) and has been used broadly in research involving latent space isotropy (5; 18; 12; 14; 15).

Mu et al. (7) build on the partition function explored by Arora et al. (23) (Equation 14), which was shown to be constant in an isotropic space over all partitions,  $c$ , where  $v_w$  is the vector representation of word  $w$ . They used the full-rank set of principal components as their partitions,  $C$ , and proposed Equation 15 as a measure of isotropy.  $I(V)$  ranges from zero to one, and holds the value one in a completely isotropic space.

$$Z_c = \sum_w \exp(c^T v_w) \quad (14)$$

$$I(V) = \frac{\min_{c \in C} Z_c}{\max_{c \in C} Z_c} \quad (15)$$

## 5.3 Measure Weaknesses

$I(V)$  and *Average Cosine Similarity* ( $E[\cos(\theta)]$ ) both have characteristic results in an isotropic space ( $E[\cos(\theta)] = 0$  and  $I(V) = 1$ ), but this does not necessarily mean that they are good measures of *relative* spread. This is highlighted by computing their values across our example distributions (Table 1).

$E[\cos(\theta)]$  appears to be particularly sensitive to the uneven spread in the *Cone* distribution, with high values across all dimension counts. However, it is insensitive to data distributions that are symmetric across the origin, even when they do not fully utilize the space. As an example, all cluster distributions produce an  $E[\cos(\theta)]$  value similar to that of the *Sphere* distribution.

$I(V)$  is also not sensitive to many of the differences in spread in our example distributions. The *Cone* distribution does have a slightly lower  $I(V)$  measure than the other three spherical distributions in low dimensions, but this becomes indistinguishable from the spherical distributions in higher dimensions. Moreover, the three cluster distributions have inconsistent results on the  $I(V)$  measure across different dimension counts. The  $I(V)$  measure is also sensitive to the random seed used to produce these distributions (data not shown).

One additional problem with  $I(V)$  is that the exponentiation in the partition function  $Z_c$  (Equation 14) causes negative projections to be heavily down-weighted when calculating the  $I(V)$  ratio. This seems a strange and undesirable behavior for a relative measure of spread, since an ideally filled space



Table 1: Example distribution results on *Average Cosine Similarity* and  $I(V)$  in 2 and 100 dimensions. Full results shown in Appendix A

| Example Distribution       | 2D                |        | 100D              |        |
|----------------------------|-------------------|--------|-------------------|--------|
|                            | $E[\cos(\theta)]$ | $I(V)$ | $E[\cos(\theta)]$ | $I(V)$ |
| <i>Shell</i>               | 0.0027            | 0.9737 | 0.0007            | 0.9988 |
| <i>Nested Shell</i>        | 0.0027            | 0.9801 | 0.0007            | 0.9982 |
| <i>Sphere</i>              | 0.0036            | 0.9613 | 0.0007            | 0.9988 |
| <i>Cone</i>                | 0.8176            | 0.9469 | 0.5119            | 0.9944 |
| <i>Symm. Clusters</i>      | 0.0020            | 0.7888 | 0.0007            | 0.9822 |
| <i>Shifted Clusters</i>    | 0.0109            | 0.9295 | 0.0012            | 0.8326 |
| <i>Uneven Clusters</i>     | 0.0249            | 0.7797 | 0.0043            | 0.9819 |
| <i>Multivariate Normal</i> | 0.0022            | 0.9969 | 0.0008            | 0.9988 |

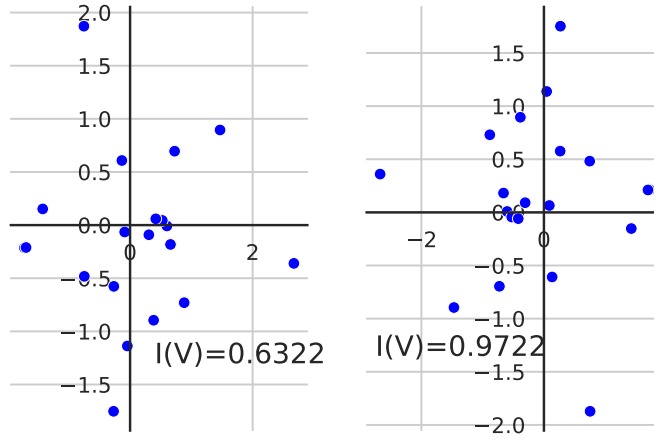


Figure 4: 2D principal component projections of identical sample data with component signs switched and  $I(V)$  calculated for each set of principal components.

must equally fill all positive and negative directions. Additionally, we note that the sign of principal components is arbitrary (24). Figure 4 demonstrates the impact of PCA’s arbitrary sign assignment on  $I(V)$ , by showing two alternative (negated) projections of randomly sampled data: when the principal components for a single distribution are negated (causing the projected data to be reflected across the origin), the resulting  $I(V)$  scores can differ greatly. We believe it is reasonable to expect a measure of spread to be constant for the same input distributions and invariant to mathematically arbitrary parameters.

## 6 Alternative Measures of Spread

To better address the issues described in Section 5.3, we explored a variety of new options for measuring spread in high dimensional space. We present two measures based on principal components/eigenvalues, one Euclidean distance measure, and four entropy-based measures.

### 6.1 Principal Component Measures

Principal components are essentially eigenvectors from a dataset’s covariance matrix, and the associated eigenvalues are what is often referred to as the *explained variance* of a principal component. In other words, the principal components (eigenvectors) describe all of the orthogonal directions for a given dataset, and the eigenvalues describe how much of the dataset’s variance can be explained along each individual direction.

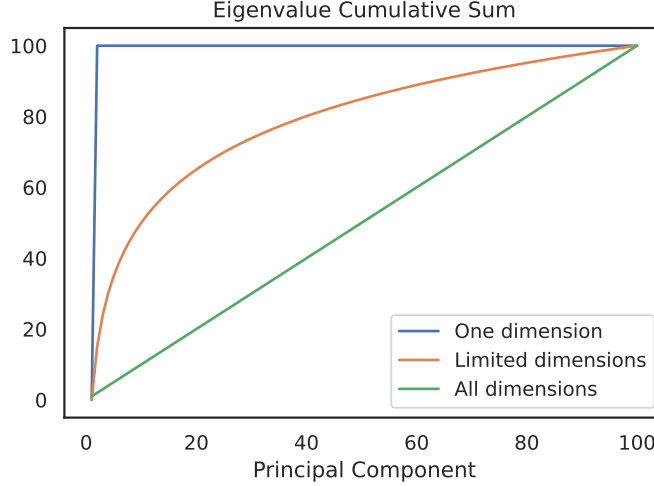


Figure 5: Cumulative Sum of Eigenvalues for latent spaces in which variance is explained on one dimension, unevenly across many dimensions, and evenly across all dimensions.

With that intuition established, we can imagine using these eigenvalues as a measure of how evenly a latent space’s variance is spread along each axis. In an ideal, well-spread space, each principal component will explain an equal amount of the dataset’s variance, resulting in all eigenvalues being roughly equal.

**Eigenvalue Ratio (ER)** A simple way to develop a measure based on this notion is to compute the ratio of the largest and smallest eigenvalues as in Equation 16, where  $C$  is the set of all  $d$  principal components in a  $d$ -dimensional space, and  $\lambda_c$  is the eigenvalue associated with component  $c$ . In an ideally filled latent space, this ratio is equal to one; if one or more dimensions is poorly used, the value will approach zero. Note that this measure provides no way to differentiate between the case of one poorly utilized dimension from the case in which many (or even most) dimensions are largely unused.

$$ER = \frac{\min_{c \in C} \lambda_c}{\max_{c \in C} \lambda_c} \quad (16)$$

**Eigenvalue Early Enrichment (EEE)** A related approach is to instead consider the cumulative sum of the eigenvalues (which are sorted largest to smallest by convention). Figure 5 provides a visual example of the approach. With an ideally filled latent space, the cumulative sum grows linearly, because each component contributes the same eigenvalue. A latent space using only one dimension will reach maximum cumulative value after the first component, so will look like an elbow. Meanwhile, a cone-shaped or otherwise unevenly filled latent space will be curved (with early eigenvalues contributing more to the sum than later eigenvalues).

This cumulative sum is converted into a relative measure by calculating the area between the cumulative sum curve and the ideal linear sum as a proportion of the total space available above that linear sum line as shown in Equation 19 (with  $v = \sum_C \lambda_c$  as the total variance in the distribution). Well-spread distributions show an *EEE* value close to zero, while poorly spread distributions approach a value of one. Unlike the previous approach, *EEE* is able to differentiate distributions that utilize different fractions of the available latent space.

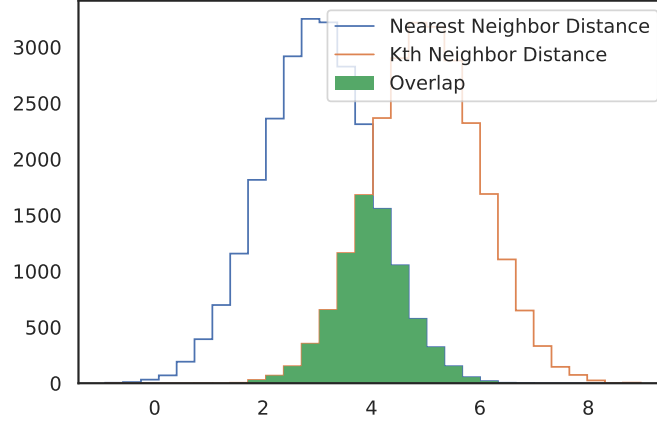


Figure 6: Based on findings from Sablayrolles et al. (17), histograms of Euclidean distance to first and  $k$ th nearest neighbors should have less overlap in a well-spread distribution than in a poorly spread distribution.

$$X_{EEE} = (x_1, x_2, \dots, x_n), \quad x_i = \sum_{c=1}^i \lambda_c \quad (17)$$

$$Y_{ref} = (y_1, y_2, \dots, y_n), \quad y_i = i \frac{v}{n} \quad (18)$$

$$EEE = \frac{AUC(X_{EEE} - Y_{ref})}{\frac{n}{2}v} \quad (19)$$

## 6.2 Euclidean Distance Measure

One (computationally expensive) way to simulate a fully used latent space is to maximize the smallest pairwise Euclidean distance between points in a distribution. Indeed, there are several works that build on this concept by creating loss functions designed to maximize the distance of each point to its nearest neighbors during training (2; 17). We sought to develop a relative measure based on this concept.

**KNN Overlap** In visualizing the effect of their nearest-neighbor-based loss function, Sablayrolles et al. (17) include a chart showing histograms for the first and 100th nearest neighbor distances for a sample of their data. The motivation behind this approach is that, for an evenly filled space, the distribution of Euclidean distances of the first nearest neighbors for all points will not overlap with (will be smaller than) the distribution of the distances to the 100th nearest neighbors. Alternatively, in an unevenly used space, these distributions will overlap. Although Sablayrolles et al. (17) did not quantify this overlap, we developed a measure based on the proportion of sampled data points falling in the intersection of the distributions for the first and  $k$ th nearest neighbor distances, where  $k = \min(5d, 100)$  (for  $d$  dimensions). This measure is visualized in Figure 6. Values range from zero (the expected value when for a well-spread distribution) to one (for a poorly spread distribution).

## 6.3 Entropy-Based Measures

In information theory, the notion of entropy quantifies the amount of uncertainty involved in predicting an outcome. When all points in a latent space map to the same location (i.e. with poorly spread data), it is easy to predict where a random point will be found – the certainty is high, and the entropy is low. In contrast, when data points are evenly spread across a latent space, it is difficult to predict the location of a randomly selected point – uncertainty is high, as is entropy. Shannon entropy is defined for continuous data as in Equation 20. In the field of astronomy, Pandey (25) proposed using Shannon entropy as a measure of the isotropy of the universe – a clear three-dimensional correlate of our goal.

$$H(p) = E(\log p(x)) = \int p(x) \log p(x) dx \quad (20)$$

For a fixed variance, this parametric definition is maximized by the normal distribution (26; 27), which matches our intuition (from Section 3.2) that a multivariate normal is a good candidate for a distribution that ideally fills the available space. Thus, the empirical Shannon entropy approximations discussed below will be compared to a multivariate normal with equal variance to create relative measures of spread.

Among our proposed entropy-based measures, two produce estimates based on distances to nearest neighbors. For the other entropy-based measures, we transformed our example distributions and multivariate normal reference data by projecting them onto their full-rank principal components before calculating entropy measures. Particularly for the proposed univariate entropy measures, this ensures that the measures are capturing information about the spread of the data without influence from any covariance between dimensions.

**Vasicek Ratio Mean Squared Error (VRM)** Our first entropy-based measure of data spread builds on a simple empirical approximation of entropy that was proposed by Vasicek in 1976 (28). The Vasicek entropy approximation is limited to univariate data and rests on the notion that ordered points will be evenly spaced in a highly entropic space. Specifically, it computes the average of log-adjusted distances between ordered data point pairs along a single dimension, considering pairs of points that are separated by a fixed interval. In Equation 21, this fixed interval,  $m$ , is a positive integer smaller than  $n/2$ , where  $n$  is the total number of data points,  $x_j = x_1$  if  $j < 1$ , and  $x_j = x_n$  if  $j > n$ . For our reference normal distribution,  $N(0, \sigma^2)$ , the Vasicek approximation converges (26) to the value presented in Equation 22.

$$H_{vas} = \frac{1}{n} \sum_{i=1}^n \log \frac{n}{2m} (x_{i+m} - x_{i-m}) \quad (21)$$

$$H_{vas}(N(0, \sigma^2)) \xrightarrow{P} \ln(\sqrt{2\pi e} \sigma^2) \quad (22)$$

To create a relative measure for data spread, we consider a ratio of the empirical value for a given distribution to the theoretical value from Equation 22. This ratio,  $J$  will equal one for normally distributed data points and will approach zero for poorly distributed data. This ratio is similar to Pielou’s evenness index from the biodiversity literature (29). As this measure applies only to univariate data, our multi-dimensional measure combines ratios over each dimension by computing the mean squared error (MSE) from the target  $J = 1$  (Equation 24).

$$J = \frac{H_{vas}}{\ln(\sqrt{2\pi e} \sigma^2)} \quad (23)$$

$$VRM = \frac{1}{d} \sum_{i=1}^d (1 - J_i)^2 \quad (24)$$

**Nearest Neighbor Entropy Ratio (NNR)** A multivariate continuous entropy approximation (27; 30; 17) rests on the notion that highly entropic spaces will maximize the distance between each point and its nearest neighbor. Equation 26 presents this nearest neighbor approximation as defined by Beirlant et al. (27), in which  $\rho_i$  is the distance of element  $i$  to its nearest neighbor, and  $e$  is the Euler constant. Since Euclidean distance can be measured in any number of dimensions, this approximation works well in high-dimensional space.<sup>5</sup>

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<sup>5</sup>To avoid the computational expense of identifying the nearest neighbor among a large number of data points, we use the FAISS library (31) to quickly identify approximate L2 nearest neighbor distances.

$$\rho_i = \min_{j \neq i} \|X_i - X_j\| \quad (25)$$

$$H_{NN} = \sum_{i=1}^n \ln(\rho_i) + \ln(2n) + e \quad (26)$$

To create a relative measure, we follow the theory behind the Clark-Evans measure of dispersion from spatial statistics by comparing this nearest-neighbor distance metric to the same metric from a known distribution (32). We simulate a  $d$ -dimensional multivariate normal distribution, with an equal number of data points to the tested distribution and fixed variance. We empirically calculate  $H_{NN}$  for this distribution as our theoretical maximum, and again use this in a ratio as in the previous section to create our proposed measure of spread. A fully used space will produce a ratio close to one and a poorly used space will produce a value approaching zero.

$$NNR = \frac{H_{NN}(X)}{H_{NN}(Y)}, \quad Y \sim N(0, \sigma^2) \quad (27)$$

**Kullback-Leibler Divergence (relative entropy)** The previous two entropy-based methods were converted into relative measures using a ratio similar to Pielou’s evenness index. Kullback-Leibler Divergence (KL-Divergence, Equation 28) provides an alternative, established framework for comparing two distributions. We compare our example distributions directly to a multivariate normal distribution to create a relative measure of spread. Below, we describe three measures of spread based on empirical KL-Divergence approximations.

$$KL = \int_x p(x) \log \frac{p(x)}{q(x)} \quad (28)$$

**Discrete KL-Divergence Mean Squared Error (DKLM)** Equation 28 is defined for continuous distributions, but KL-Divergence can be computed for finite data sets by discretizing the data as in Equation 29. For this measure, we estimate univariate  $p(X')$  and  $q(Y')$  by binning observed and reference data  $(X, Y)$  into  $k$  bins along each dimension, with  $k = 30$ .

$$KL_{disc} = \sum_{i=1}^k p(x'_i) \log \frac{p(x'_i)}{q(x'_i)} \quad (29)$$

To produce a relative measure of spread in high-dimensional space, we again join these univariate measures by computing the mean squared difference from a reference of  $KL_{disc} = 0$  over all  $d$  dimensions, as in Equation 30.

$$DKLM = \frac{1}{d} \sum_{i=1}^d (KL_{disc_i})^2 \quad (30)$$

**Gaussian KL-Divergence (GKL)** Another common method for calculating KL-Divergence is to estimate continuous parameters from the observed distribution and then calculate the closed-form of Equation 28. Here,  $\mu_p$  and  $\sigma_p$  are the mean and covariance matrix of the observed distribution, and  $d$  is the number of dimensions. We propose a measure based on the closed-form KL-divergence between two multivariate Gaussians in Equation 31, where the  $tr()$  function is the sum of the diagonal elements of the matrix (33). Although we don’t expect our example distributions to be well-approximated by a normal distribution, this method shows up frequently in machine learning frameworks (e.g. Variational Autoencoders (34), t-SNE (35), Reinforcement Learning (36)).

$$GKL = \frac{1}{2} (\mu_p^\top \mu_p + tr(\sigma_p) - d - \log |\sigma_p|) \quad (31)$$

**Nearest Neighbor KL-Divergence (NNKL)** Finally, we propose a KL-Divergence approximation based on comparing the distributions of nearest neighbor distances in our observed and reference distributions, following Pérez-Cruz (37). Equations 32 and 33 approximate the probability distribution in our observed and reference distributions, with  $s(x_i)$  and  $r(x_i)$  being the nearest neighbor distance to point  $x_i$  in our observed and reference distributions, respectively. Substituting these into a modified KL-divergence formula results in Equation 34 (see Pérez-Cruz (37) for derivation), which relies only on these nearest neighbor distances, the number of dimensions,  $d$ , and the number of data points,  $n$  (observed) and  $m$  (reference).

$$\hat{p}(x_i) = \frac{1}{(n-1)} \frac{\Gamma(n/2 + 1)}{\pi^{n/2} s(x_i)^n} \quad (32)$$

$$\hat{q}(x_i) = \frac{1}{m} \frac{\Gamma(n/2 + 1)}{\pi^{n/2} r(x_i)^n} \quad (33)$$

$$NNKL = \frac{1}{n} \sum_{i=1}^n \log \frac{\hat{p}(x_i)}{\hat{q}(x_i)} = \frac{d}{n} \sum_{i=1}^n \log \frac{r(x_i)}{s(x_i)} + \log \frac{m}{n-1} \quad (34)$$

By design, our observed and reference distributions always have the same number of data points, so we ignore the final term in Equation 34. Additionally, we observe that the equation will “credit” data points for which  $r_k(x_i)$  is smaller than  $s_k(x_i)$ , because this contributes a negative value to the  $NNKL$  sum, making a negative KL-Divergence possible. For this reason, we limit the contribution of any given data point to zero, yielding the final Equation (35) for our proposed *Nearest-Neighbor KL-Divergence* approximation.

$$NNKL = \frac{d}{n} \sum_{i=1}^n \max(0, \log \frac{r_k(x_i)}{s_k(x_i)}) \quad (35)$$

## 7 Results

In Section 4, we introduced seven example distributions with varying expected relative spread. In Section 5, we showed that the commonly used isotropy measures *Average Cosine Similarity* and  $I(V)$  do not match the expectations for a nuanced measure of spread. Here, we evaluate each of the alternative measures proposed in Section 6, and demonstrate that all measures except *KNN Overlap* better reflect our expectations as relative measures of spread, particularly for our cluster-based distributions in high dimensions. We will review the results for each of our proposed measures below<sup>6</sup>.

### 7.1 Principal Component Measures

Our proposed *Eigenvalue Ratio* (ER) measure considers the ratio between the largest and smallest eigenvalues of a data distribution (i.e. the explained variance for the first and last principal component), where a ratio of one indicates a well-spread distribution (Equation 16). The *Early Eigenvalue Enrichment* (EEE) score instead considers the area under the curve of the cumulative sum of eigenvalues as a proportion of the total area available, where a score of zero indicates a well-spread distribution (Equation 19).

Both of these measures behave as we would expect for relative measures of spread in our example distributions across all number of dimensions. As seen in Table 2, both measures show a gradient of spread across our example distributions in 2 and 100 dimensions, with the *Sphere* and *Shell* distributions most evenly spread and the three cluster distributions least well-spread.

Although both of these measures seem to perform well on our example distributions, the EEE score better agrees with our expectations for a relative measure of spread, since it considers more than just the first and last eigenvalue. Indeed, Table 2 shows a disagreement between the ER score and the

<sup>6</sup>Patterns described in the main body hold for 10 and 50 dimensions as well, and complete results can be found in Appendix A.

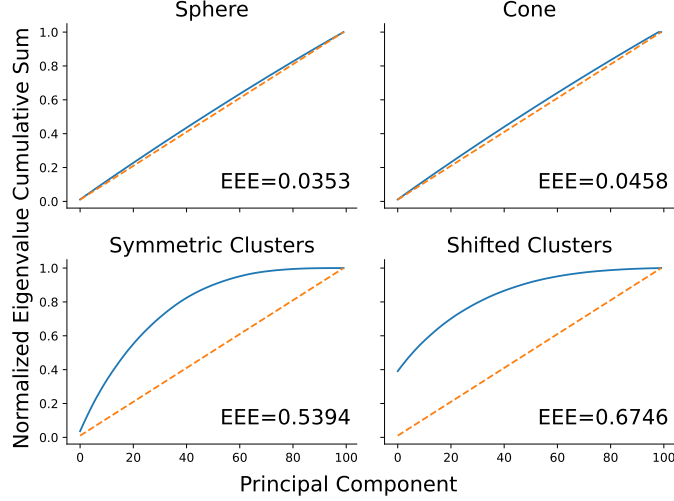


Figure 7: *Eigenvalue Early Enrichment (EEE)* scores for *Sphere*, *Cone*, *Symmetric Clusters*, and *Shifted Clusters* distributions

Table 2: Example distribution results on *Eigenvalue Ratio (ER)* and *Eigenvalue Early Enrichment (EEE)* in 2 and 100 dimensions.

| Example Distribution       | 2D     |        | 100D   |        |
|----------------------------|--------|--------|--------|--------|
|                            | ER     | EEE    | ER     | EEE    |
| <i>Shell</i>               | 0.9021 | 0.0129 | 0.7845 | 0.0361 |
| <i>Nested Shell</i>        | 0.8552 | 0.0195 | 0.7542 | 0.0422 |
| <i>Sphere</i>              | 0.9118 | 0.0115 | 0.7889 | 0.0353 |
| <i>Cone</i>                | 0.4173 | 0.1028 | 0.0090 | 0.0458 |
| <i>Symm. Clusters</i>      | 0.1124 | 0.1995 | 0.0001 | 0.5394 |
| <i>Shifted Clusters</i>    | 0.0915 | 0.2081 | 0.0007 | 0.6746 |
| <i>Uneven Clusters</i>     | 0.1102 | 0.2004 | 0.0001 | 0.5397 |
| <i>Multivariate Normal</i> | 0.7910 | 0.0292 | 0.7804 | 0.0362 |

EEE score on the 100-dimensional cluster distributions, where the ER scores suggests that the *Shifted Clusters* distribution has more complete spread, while the EEE scores suggests that the *Symmetrical Clusters* distribution has more complete spread. This difference can be explained by examining the eigenvalue cumulative sum curves in Figure 7: the ER score penalizes the *Symmetric Clusters* distribution for having a very small last eigenvalue, while the EEE score more accurately reflects the more gradual decrease over all eigenvalues when compared to the *Shifted Clusters* distribution.

## 7.2 Euclidean Distance Measure

The *KNN Overlap* score considers the proportion of a data distribution with overlapping first and  $k$ th nearest neighbor distances, where a well-spread distribution is expected to have no overlapping distances (Figure 6). This was the only one of our proposed measures that didn't clearly improve on commonly used measures. As seen in Table 3, this measure is very small for almost all example distributions. And although the 2-dimensional distribution values reflect a reasonable relative measure of spread, these relative differences become smaller as the number of dimensions increases. We did try adjusting the value for  $k$  to account for this difference, but found that the scores for the cluster distributions were particularly sensitive to the choice of  $k$ .

Figure 8 shows that the *Uneven Clusters* distribution has very high overlap with  $k = 100$ . In the *Uneven Clusters* distribution, the randomization of cluster size will cause many of the clusters to have more than the standard 125 data points used in the *Symmetric Clusters* distribution. However, the

Table 3: Example distribution results for *KNN Overlap* in 2 and 100 dimensions.

| Example Distribution       | 2D     | 100D   |
|----------------------------|--------|--------|
| <i>Shell</i>               | 0.0120 | 0.0000 |
| <i>Nested Shell</i>        | 0.0480 | 0.0000 |
| <i>Sphere</i>              | 0.0240 | 0.0001 |
| <i>Cone</i>                | 0.0100 | 0.0001 |
| <i>Symm. Clusters</i>      | 0.1560 | 0.0078 |
| <i>Shifted Clusters</i>    | 0.1700 | 0.0067 |
| <i>Uneven Clusters</i>     | 0.1960 | 0.8491 |
| <i>Multivariate Normal</i> | 0.1960 | 0.2260 |

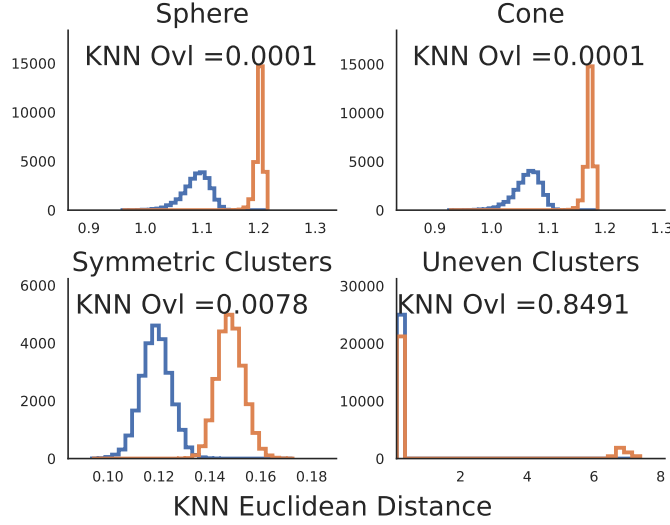


Figure 8: KNN Overlap Proportion for 100D *Sphere*, *Cone*, *Symmetric Clusters*, and *Shifted Clusters* distributions, with the nearest neighbor distances in blue, and the 100th nearest neighbor distances in orange

randomly sized clusters still share the same tight distribution around a point, causing the 100th nearest neighbor to very frequently be quite close. Alternatively, increasing  $k$  causes the *KNN Overlap* score for the *Symmetric Clusters* and *Shifted Clusters* distributions to drop to zero, since their clusters are all exactly 125 points (meaning that a point’s 126th nearest neighbor is almost guaranteed to be much farther away than its first nearest neighbor).

### 7.3 Entropy-Based Measures

We proposed two univariate entropy measures of spread, *Vasicek Ratio MSE (VRM)* and *Discrete KL-Divergence MSE (DKLM)*. VRM considers the ratio of the Vasicek entropy approximation between the observed distribution and a multivariate normal distribution (Equation 21). DKLM considers the KL-divergence between a discretized observed distribution and a multivariate normal distribution (Equation 29). As both of these entropy measures are univariate, we proposed to use the mean squared error over all dimensions as our multivariate measures of spread, and thus zero indicates a well-spread distribution for both measures.

Both VRM and DKLM behave as we would expect for quality relative measures of spread, particularly in high dimensions (see Table 4). Both measures produce small MSE for the *Sphere*, *Shell*, and *Nested Shell* distributions, but a larger MSE for the *Cone* distribution. The cluster distributions result in even larger MSE, which fits with our intuition about how these distributions should perform in terms of their relative spread.



Table 4: Example distribution results for *Vasicek Ratio MSE (VRM)* and *Discrete KL-Divergence MSE (DKLM)* in 2 and 100 dimensions.

| Example Distribution       | 2D     |        | 100D   |         |
|----------------------------|--------|--------|--------|---------|
|                            | VRM    | DKLM   | VRM    | DKLM    |
| <i>Shell</i>               | 0.1891 | 0.1349 | 0.0010 | 0.0000  |
| <i>Nested Shell</i>        | 0.0465 | 0.0774 | 0.0014 | 0.0019  |
| <i>Sphere</i>              | 0.0106 | 0.0169 | 0.0009 | 0.0000  |
| <i>Cone</i>                | 0.0686 | 0.0266 | 0.0097 | 0.0295  |
| <i>Symm. Clusters</i>      | 0.1545 | 0.1434 | 0.2624 | 0.6200  |
| <i>Shifted Clusters</i>    | 0.2143 | 0.5392 | 0.2386 | 20.5810 |
| <i>Uneven Clusters</i>     | 0.1637 | 0.1470 | 0.2574 | 0.6333  |
| <i>Multivariate Normal</i> | 0.0047 | 0.0190 | 0.0010 | 0.0000  |

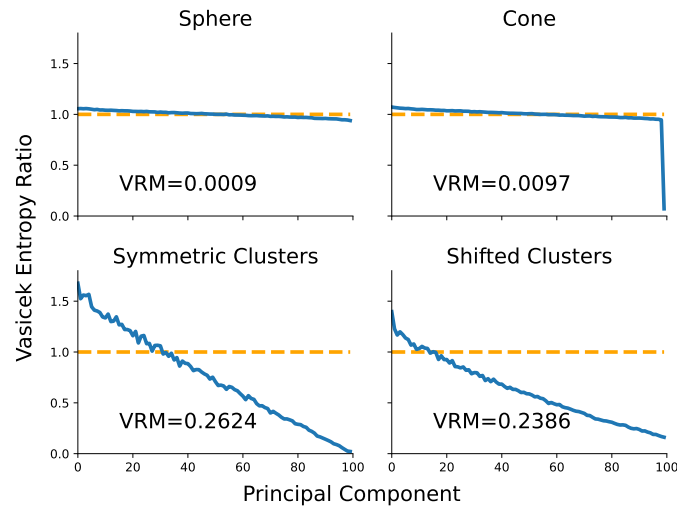


Figure 9: *Vasicek Entropy Ratio MSE (VRM)* for 100-dimensional *Sphere*, *Cone*, *Symmetric Clusters*, and *Shifted Clusters* distributions. A well-spread distribution will have a ratio of one over all dimensions, resulting in an MSE of zero.

Figure 9 shows the *Vasicek Entropy Ratio* over all dimensions in blue, with the reference line in orange. In a well-spread distribution, the ratio would always be 1 and the MSE from the reference line would be zero.

The consistently negative slope on these plots is an effect of projecting our data onto their full-rank principal components. By convention, principal components are in descending order of their explained variance, and components with more variance (early components) will have a higher *Vasicek Entropy Ratio* than those with very little variance (late components).

Figure 10 shows the *Discrete KL-Divergence* over all dimensions. Here a well-spread distribution would have a KL-divergence of zero, resulting in an MSE of zero. These plots show how the same pattern of decreasing variance over the principal components affects KL-Divergence, where a difference in variance causes an increase in KL-Divergence, regardless of whether there is more or less variance than in the reference data.

One issue with all of our KL-Divergence measures is that KL-Divergence does not have an upper limit. The *Shifted Clusters* chart in Figure 10 shows how just a few components dominate the DKLM score for this distribution, which would not be as pronounced in a bounded measure. Indeed, our empirical KL-divergence approximations are functions of the number of data points and the number of dimensions, such that the range of scores for distributions in high dimensions is much larger than

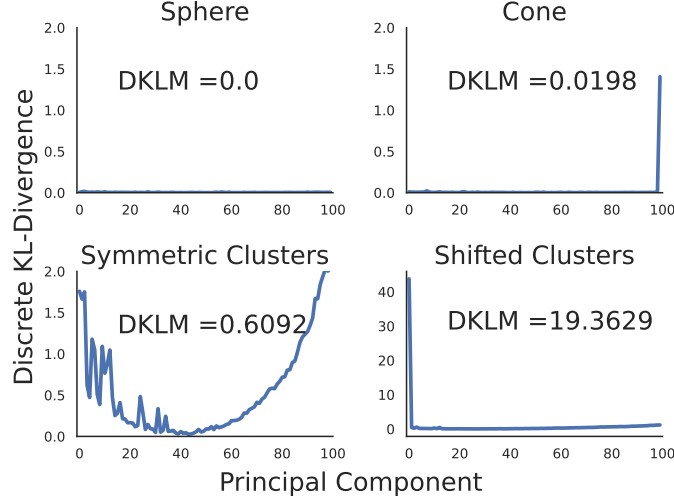


Figure 10: *Discrete KL-Divergence MSE (DKLM)* for 100-dimensional *Sphere*, *Cone*, *Symmetric Clusters*, and *Shifted Clusters* distributions. A well-spread distribution will have a KL-divergence of zero over all dimensions, resulting in an MSE of zero.

Table 5: Example distribution results for *Gaussian KL-Divergence (GKL)* in 2 and 100 dimensions.

| Example Distribution       | 2D     | 100D    |
|----------------------------|--------|---------|
| <i>Shell</i>               | 0.0013 | 0.0985  |
| <i>Nested Shell</i>        | 0.0031 | 0.1344  |
| <i>Sphere</i>              | 0.0011 | 0.0945  |
| <i>Cone</i>                | 0.0926 | 1.8939  |
| <i>Symm. Clusters</i>      | 0.5064 | 18.4996 |
| <i>Shifted Clusters</i>    | 0.5902 | 18.4996 |
| <i>Uneven Clusters</i>     | 0.5141 | 18.4996 |
| <i>Multivariate Normal</i> | 0.0069 | 0.0993  |

in lower dimensional distributions. Ideally, a reliable measure of spread would be agnostic to the size or number of dimensions in any observed distribution, allowing meaningful comparison between many different distributions.

The proposed *Gaussian KL-Divergence (GKL)* measure of spread computes the closed-form KL-Divergence between a multivariate normal distribution and a Gaussian approximation of the observed data (Equation 31). In a well-spread distribution, we would expect the KL-divergence from a multivariate normal to be zero.

Given that we did not expect our example distributions to be well-modeled by Gaussians, we were surprised to find that Table 5 shows the same pattern of effectiveness in measuring relative spread for GKL. Again, this measure indicates that data spread becomes less complete as we move from the three spherical distributions, to the *Cone* distribution, and to the cluster distributions. However, this measure also suffers from the weakness of KL-Divergence measures described above, in that the range for high dimensional distributions is much larger than it is for lower dimensional distributions.

Finally, we consider our two proposed nearest-neighbor entropy measures. *Nearest Neighbor Entropy Ratio (NNR)* considers a ratio between the nearest-neighbor entropy approximation of the observed data and that of a multivariate normal distribution (Equation 26). *Nearest Neighbor KL-Divergence (NNKL)* considers a nearest-neighbor approximation of the KL-divergence between the observed data and a multivariate normal distribution. In a well-spread distribution, we would expect an *NNR* of one, and an *NNKL* of zero.

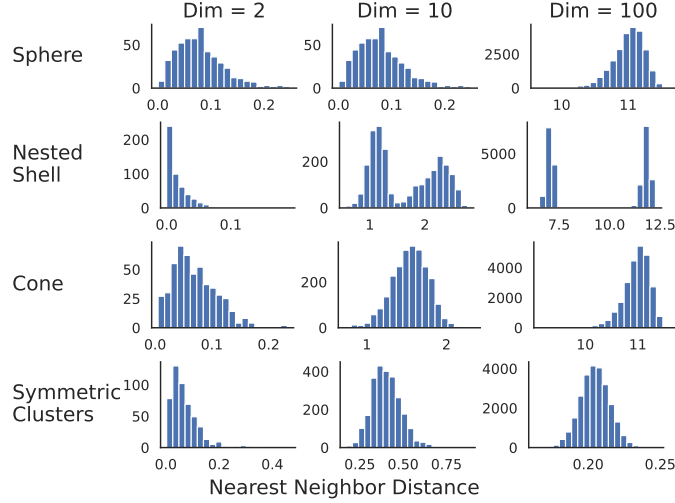


Figure 11: Histograms of nearest neighbor distances for our *Sphere*, *Nested Shell*, *Cone*, and *Symmetric Clusters* example distributions in 2, 10, and 100 dimensions.

Table 6 shows that, although these measures are once again effective at identifying the less-complete spread of the cluster distributions, the *Cone* distribution receives similar scores to the *Sphere* and *Shell* distributions, while the scores for the *Nested Shell* distribution indicate a less complete spread, especially in high dimensions. This effect can be explained by considering histograms of our example distributions’ nearest neighbor distances in Figure 11. These plots show that nearest neighbor distances for the *Sphere* and *Cone* distributions are very similar as the number of dimensions increases, while the nearest neighbor distances for the *Nested Shell* distribution become increasingly separated for the two “rings” in the distribution. In the *Symmetric Clusters* distribution (and indeed all of the cluster distributions), the nearest neighbor distances change much less as the number of dimensions increase.

Table 6 also provides the strongest example of the issue raised above for KL-Divergence, with the NNKL scores for the 100-dimensional cluster distributions being almost 300x the scores of their 2-dimensional counterparts.

Table 6: Example distribution results on *Nearest Neighbor Entropy Ratio (NNR)* and *Nearest Neighbor KL-Divergence (NNKL)* in 2 and 100 dimensions.

| Example Distribution       | 2D     |        | 100D   |          |
|----------------------------|--------|--------|--------|----------|
|                            | NNR    | NNKL   | NNR    | NNKL     |
| <i>Shell</i>               | 0.1323 | 7.4573 | 1.0024 | 0.2532   |
| <i>Nested Shell</i>        | 0.3996 | 5.2696 | 0.9887 | 13.2125  |
| <i>Sphere</i>              | 0.9749 | 0.9267 | 1.0023 | 0.2558   |
| <i>Cone</i>                | 0.9350 | 1.2659 | 1.0022 | 0.2853   |
| <i>Symm. Clusters</i>      | 0.8971 | 1.4874 | 0.7135 | 394.5467 |
| <i>Shifted Clusters</i>    | 0.8312 | 1.9226 | 0.6882 | 429.2692 |
| <i>Uneven Clusters</i>     | 0.9015 | 1.4215 | 0.7134 | 394.7038 |
| <i>Multivariate Normal</i> | 0.9973 | 0.7260 | 1.0001 | 1.1751   |

## 7.4 Word2Vec Embedding Performance

The work in this paper is largely motivated by creating better metrics for comparing the spread in natural language processing models’ latent spaces. Thus, as a final exploration, we compare the performance of a pre-trained Word2Vec model (300-dimensions trained on the Google News corpus,

Mikolov (38)) on our two best candidate measures, *Early Eigenvalue Enrichment (EEE)* and *Vasicek Ratio MSE (VRM)*, along with the current common measures, *Average Cosine Similarity* and *I(V)*. We follow the methods used in generating our example distributions and randomly sample 75000 (250d) word embeddings from this pre-trained model to use in calculating our measures of data spread. To examine whether these measures can capture a gradual increase in data spread when applied to a real (not simulated) latent space, we add random uniform noise to the pre-trained embeddings and calculate the *EEE*, *VRM*, *Average Cosine Similarity*, and *I(V)* scores at each addition.

Table 7 shows that the original pre-trained Word2Vec model falls somewhere between our example *Cone* distribution and our *Symmetric Clusters* distribution according to all four of these measures of spread. Additionally, we see that all four measures successfully reflect the gradual increase in data spread as random noise is added to the pre-trained word embeddings. This does not discount the issues that we raised in Section 5, as we’ve demonstrated that the *Average Cosine Similarity* and *I(V)* scores misbehave with poorly spread distributions that have certain characteristics, such as maintaining symmetry across the origin. However, this examination further supports the use of *EEE* and *VRM* as relative measures of data spread that do not suffer from the same weaknesses.

Table 7: Pre-trained Word2Vec Results on *Early Eigenvalue Enrichment (EEE)*, *Vasicek Ratio MSE (VRM)*, *Average Cosine Similarity*, and *I(V)* measures with gradual addition of random uniform noise.

| Uniform Noise Range | EEE    | VRM    | $E[\cos(\theta)]$ | I(V)   |
|---------------------|--------|--------|-------------------|--------|
| <i>None</i>         | 0.4058 | 0.1525 | 0.1317            | 0.9251 |
| (-0.001, 0.001)     | 0.4058 | 0.1525 | 0.1319            | 0.9251 |
| (-0.01, 0.01)       | 0.4049 | 0.1514 | 0.1280            | 0.9257 |
| (-0.05, 0.05)       | 0.3861 | 0.1320 | 0.1147            | 0.9359 |
| (-0.1, 0.1)         | 0.3375 | 0.0961 | 0.0933            | 0.9534 |
| (-0.3, 0.3)         | 0.1492 | 0.0216 | 0.0346            | 0.9883 |
| (-0.5, 0.5)         | 0.0800 | 0.0066 | 0.0192            | 0.9952 |
| (-1, 1)             | 0.0441 | 0.0016 | 0.0062            | 0.9991 |
| (-3, 3)             | 0.0367 | 0.0010 | 0.0022            | 0.9996 |

## 8 Conclusion

In this work, we have examined methods for quantifying how completely data fills a latent space. This has become a driving factor in much work around improving and optimizing natural language processing models. While there has been quite a bit of research around increasing the usage of a model’s latent space, we demonstrated that the metrics being used to quantify this usage appear to be insufficient as relative measures of data spread, particularly in high dimensional latent spaces.

We designed seven example distributions based on geometric shapes or patterns for which we have some intuition about how their relative data spread should compare, and reproduced these distribution shapes in 2, 10, 50, and 100 dimensions. We then proposed and evaluated eight new measures of spread and compared these measures with two commonly used measures across all seven example distributions. These comparisons allowed us to explore options for a more accurate way of quantifying how completely a latent space is filled.

Looking at the two commonly used measures of data spread, we found that neither behaves as we would expect for a relative measure of data spread on our example distributions. *Average Cosine Similarity* is sensitive to distributions that are not symmetrical around the origin, but fails to identify symmetrically incomplete patterns of data spread. *I(V)*, a measure based on the magnitude of principal component projections, was not sensitive to any of our pathological example distributions, particularly in higher dimensions. Additionally, *I(V)* is greatly affected by the sign of the principal components, which is known to be arbitrary, providing more evidence against *I(V)* as a reliable measure of data spread for comparison between latent spaces.

Our proposed measures included two principal component based measures, the *Eigenvalue Ratio (ER)* and the *Eigenvalue Early Enrichment (EEE)* scores. Both of these measures performed well as relative measures of spread, following our intuition on how the example distributions should score

across all numbers of dimensions. However, of the two the *EEE* score is a more reliable measure since it is less sensitive to an especially small last eigenvalue, and it produces scores that are more similar (and thus more comparable) across different numbers of dimensions.

The one proposed measure that failed to improve upon those currently used in the literature was *KNN Overlap* which is computed based on the proportion of overlap between the distributions of distances between points to their first and  $k$ th nearest neighbor. We found that this measure varies greatly based on the choice of  $k$ , and finding a  $k$  that is sensitive to differences in spread across many different distributions would be very difficult, if not impossible.

We also considered several entropy-based measures of data spread, and once again found that all of the measures proposed behaved more reliably than commonly used measures on our example distributions. However, we found that the three proposed KL-Divergence measures (*Gaussian KL-Divergence (GKL)*, *Discrete KL-Divergence MSE (DKLM)*, and *Nearest Neighbor KL-Divergence (NNKL)*) were too closely linked with the number of dimensions to be ideal relative measures of spread. We also found that the *Nearest Neighbor Entropy Ratio (NNR)* did not quite follow our intuition on some of our example distributions, due to relying solely on nearest neighbor distances as a proxy for data spread. As a result of this evaluation, we recommend the *Vasicek Ratio MSE (VRM)* score for a reliable entropy-based measure of data spread, as it performs according to our intuition and results in similar, comparable scores between distributions and different numbers of dimensions.

## Acknowledgments and Disclosure of Funding

We thank Doug Brinkerhoff, John Chandler, and Jon Graham for sharing valuable insights during formative stages of this exploration. Without their engagement, this would be a lesser work.

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## A Extended Results

Table 8: Example distribution results on *Average Cosine Similarity* and  $I(V)$ .

| Example Distribution       | 2D                |        | 10D               |        | 50D               |        | 100D              |        |
|----------------------------|-------------------|--------|-------------------|--------|-------------------|--------|-------------------|--------|
|                            | $E[\cos(\theta)]$ | $I(V)$ | $E[\cos(\theta)]$ | $I(V)$ | $E[\cos(\theta)]$ | $I(V)$ | $E[\cos(\theta)]$ | $I(V)$ |
| <i>Shell</i>               | 0.0027            | 0.9737 | 0.0008            | 0.9916 | 0.0008            | 0.9977 | 0.0007            | 0.9988 |
| <i>Nested Shell</i>        | 0.0027            | 0.9801 | 0.0008            | 0.9881 | 0.0008            | 0.9964 | 0.0007            | 0.9982 |
| <i>Sphere</i>              | 0.0036            | 0.9613 | 0.001             | 0.9905 | 0.0008            | 0.9976 | 0.0007            | 0.9988 |
| <i>Cone</i>                | 0.8176            | 0.9469 | 0.5728            | 0.9471 | 0.5201            | 0.9890 | 0.5119            | 0.9944 |
| <i>Symm. Clusters</i>      | 0.0020            | 0.7888 | 0.0004            | 0.8857 | 0.0007            | 0.9643 | 0.0007            | 0.9822 |
| <i>Shifted Clusters</i>    | 0.0109            | 0.9295 | 0.0108            | 0.8251 | 0.0023            | 0.8360 | 0.0012            | 0.8326 |
| <i>Uneven Clusters</i>     | 0.0249            | 0.7797 | 0.0183            | 0.8972 | 0.0076            | 0.9664 | 0.0043            | 0.9819 |
| <i>Multivariate Normal</i> | 0.0022            | 0.9969 | 0.0007            | 0.9934 | 0.0008            | 0.9974 | 0.0008            | 0.9988 |

Table 9: Example distribution results on *Eigenvalue Ratio (ER)* and *Eigenvalue Early Enrichment (EEE)*.

| Example Distribution       | 2D     |        | 10D    |        | 50D    |        | 100D   |        |
|----------------------------|--------|--------|--------|--------|--------|--------|--------|--------|
|                            | ER     | EEE    | ER     | EEE    | ER     | EEE    | ER     | EEE    |
| <i>Shell</i>               | 0.9021 | 0.0129 | 0.8380 | 0.0307 | 0.7943 | 0.0354 | 0.7845 | 0.0361 |
| <i>Nested Shell</i>        | 0.8552 | 0.0195 | 0.7991 | 0.0381 | 0.7550 | 0.0417 | 0.7542 | 0.0422 |
| <i>Sphere</i>              | 0.9118 | 0.0115 | 0.8210 | 0.0323 | 0.7845 | 0.0363 | 0.7889 | 0.0353 |
| <i>Cone</i>                | 0.4173 | 0.1028 | 0.0807 | 0.1155 | 0.0185 | 0.0536 | 0.0090 | 0.0458 |
| <i>Symm. Clusters</i>      | 0.1124 | 0.1995 | 0.0126 | 0.4987 | 0.0003 | 0.5503 | 0.0001 | 0.5394 |
| <i>Shifted Clusters</i>    | 0.0915 | 0.2081 | 0.0070 | 0.5792 | 0.0021 | 0.6558 | 0.0007 | 0.6746 |
| <i>Uneven Clusters</i>     | 0.1102 | 0.2004 | 0.0134 | 0.5067 | 0.0003 | 0.5506 | 0.0001 | 0.5397 |
| <i>Multivariate Normal</i> | 0.7910 | 0.0292 | 0.7917 | 0.0398 | 0.7839 | 0.0368 | 0.7804 | 0.0362 |

Table 10: Example distribution results for *KNN Overlap*.

| Example Distribution       | 2D     | 10D    | 50D    | 100D   |
|----------------------------|--------|--------|--------|--------|
| <i>Shell</i>               | 0.0120 | 0.0000 | 0.0000 | 0.0000 |
| <i>Nested Shell</i>        | 0.0480 | 0.0924 | 0.0000 | 0.0000 |
| <i>Sphere</i>              | 0.0240 | 0.0000 | 0.0002 | 0.0001 |
| <i>Cone</i>                | 0.0100 | 0.0008 | 0.0004 | 0.0001 |
| <i>Symm. Clusters</i>      | 0.1560 | 0.0508 | 0.0061 | 0.0078 |
| <i>Shifted Clusters</i>    | 0.1700 | 0.0516 | 0.0078 | 0.0067 |
| <i>Uneven Clusters</i>     | 0.1960 | 0.0896 | 0.8468 | 0.8491 |
| <i>Multivariate Normal</i> | 0.1960 | 0.1580 | 0.1940 | 0.2260 |

Table 11: Example distribution results for *Vasicek Ratio MSE (VRM)* and *Discrete KL-Divergence MSE (DKLM)*.

| Example Distribution       | 2D     |        | 10D    |        | 50D    |         | 100D   |         |
|----------------------------|--------|--------|--------|--------|--------|---------|--------|---------|
|                            | VRM    | DKLM   | VRM    | DKLM   | VRM    | DKLM    | VRM    | DKLM    |
| <i>Shell</i>               | 0.1891 | 0.1349 | 0.0012 | 0.0003 | 0.0010 | 0.0000  | 0.0010 | 0.0000  |
| <i>Nested Shell</i>        | 0.0465 | 0.0774 | 0.0013 | 0.0027 | 0.0014 | 0.0025  | 0.0014 | 0.0019  |
| <i>Sphere</i>              | 0.0106 | 0.0169 | 0.0011 | 0.0003 | 0.001  | 0.0000  | 0.0009 | 0.0000  |
| <i>Cone</i>                | 0.0686 | 0.0266 | 0.0603 | 0.0735 | 0.0172 | 0.0521  | 0.0097 | 0.0295  |
| <i>Symm. Clusters</i>      | 0.1545 | 0.1434 | 0.1983 | 0.6150 | 0.2655 | 0.7538  | 0.2624 | 0.6200  |
| <i>Shifted Clusters</i>    | 0.2143 | 0.5392 | 0.2401 | 0.6748 | 0.2398 | 22.5188 | 0.2386 | 20.5810 |
| <i>Uneven Clusters</i>     | 0.1637 | 0.1470 | 0.1990 | 0.8154 | 0.2637 | 0.8012  | 0.2574 | 0.6333  |
| <i>Multivariate Normal</i> | 0.0047 | 0.0190 | 0.0012 | 0.0017 | 0.0010 | 0.0001  | 0.0010 | 0.0000  |

Table 12: Example distribution results for *Gaussian KL-Divergence (GKL)*.

| Example Distribution       | 2D     | 10D    | 50D     | 100D    |
|----------------------------|--------|--------|---------|---------|
| <i>Shell</i>               | 0.0013 | 0.0078 | 0.0478  | 0.0985  |
| <i>Nested Shell</i>        | 0.0031 | 0.0122 | 0.0662  | 0.1344  |
| <i>Sphere</i>              | 0.0011 | 0.0086 | 0.0500  | 0.0945  |
| <i>Cone</i>                | 0.0926 | 0.7479 | 1.4945  | 1.8939  |
| <i>Symm. Clusters</i>      | 0.5064 | 3.5500 | 18.5039 | 18.4996 |
| <i>Shifted Clusters</i>    | 0.5902 | 4.3510 | 18.5038 | 18.4996 |
| <i>Uneven Clusters</i>     | 0.5141 | 3.6196 | 18.5039 | 18.4996 |
| <i>Multivariate Normal</i> | 0.0069 | 0.0130 | 0.0516  | 0.0993  |

Table 13: Example distribution results on *Nearest Neighbor Entropy Ratio (NNR)* and *Nearest Neighbor KL-Divergence (NNKL)*.

| Example Distribution       | 2D     |        | 10D    |         | 50D    |          | 100D   |          |
|----------------------------|--------|--------|--------|---------|--------|----------|--------|----------|
|                            | NNR    | NNKL   | NNR    | NNKL    | NNR    | NNKL     | NNR    | NNKL     |
| <i>Shell</i>               | 0.1323 | 7.4573 | 0.9981 | 0.8837  | 1.0028 | 0.3357   | 1.0024 | 0.2532   |
| <i>Nested Shell</i>        | 0.3996 | 5.2696 | 0.9950 | 1.0005  | 0.9899 | 5.3053   | 0.9887 | 13.2125  |
| <i>Sphere</i>              | 0.9749 | 0.9267 | 0.9994 | 0.8082  | 1.0026 | 0.3703   | 1.0023 | 0.2558   |
| <i>Cone</i>                | 0.9350 | 1.2659 | 0.9939 | 1.1920  | 1.0022 | 0.4667   | 1.0022 | 0.2853   |
| <i>Symm. Clusters</i>      | 0.8971 | 1.4874 | 0.8505 | 14.1981 | 0.7458 | 160.3722 | 0.7135 | 394.5467 |
| <i>Shifted Clusters</i>    | 0.8312 | 1.9226 | 0.8111 | 17.5756 | 0.7188 | 177.1980 | 0.6882 | 429.2692 |
| <i>Uneven Clusters</i>     | 0.9015 | 1.4215 | 0.8495 | 13.9207 | 0.7457 | 160.4552 | 0.7134 | 394.7038 |
| <i>Multivariate Normal</i> | 0.9973 | 0.7260 | 0.9996 | 0.7083  | 0.9999 | 0.9290   | 1.0001 | 1.1751   |